ZynerjiChirality v0.7.0

Differential Spectral Fingerprints and Bridged Ring Conformer Improvements

Technical Whitepaper

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v0.7.0 Key Metrics

Metric	**Value**
Differential FP enantiomer cosine	-0.09 to -0.93 (anti-correlated)
Full FP enantiomer cosine (v0.6.0)	0.72–0.98 (topology-dominated)
Achiral differential FP norm	≈ 0
Bridged ring embed success	100% (norbornane, camphor, DXM)
Core detection accuracy	100% (amino acids, drugs, achiral)
Per-target models trained	986/986
Total tests passing	296

Abstract

ZynerjiChirality v0.7.0 introduces the **chirality differential fingerprint**, a 128-dimensional spectral embedding that isolates the chirality signal from shared molecular topology. The v0.6.0 full fingerprint concatenated 8 sub-vectors ($8k$ raw dimensions), of which 6 encode topology shared between enantiomers, resulting in cosine similarities of 0.87–0.99 for enantiomer pairs—far from the theoretical -1.0. The differential fingerprint uses only the 2 chirality-sensitive sub-vectors (diff_cos and diff_sin), which flip sign between enantiomers by construction, yielding cosine < 0.5 and near-zero norm for achiral molecules.

Additionally, v0.7.0 fixes conformer embedding for bridged bicyclic molecules (norbornane, camphor, dextromethorphan) via a dedicated fallback chain with MMFF optimization at all stages, and exposes both fingerprint types directly on the `ChiralityResult` object for downstream ML pipelines.

Contents

1 Introduction

ZynerjiChirality detects molecular chirality via **dual-helix spectral graph analysis**: the eigenvalue spectra of two phase-modulated graph Laplacians (cos and sin helices with golden-ratio coupling) respond differently to R vs. S stereochemistry when the adjacency matrix is ordered by CIP priority. The chirality signal appears as the *differential asymmetry*—the change in the cos–sin eigenvalue gap between chirality-modulated and standard (unmodulated) adjacency matrices.

1.1 Motivation for v0.7.0

The v5 Atlas benchmark (Section **??**) revealed two limitations:

1. **Topology dilution**: The 128-dim spectral fingerprint from v0.6.0 concatenates 8 sub-vectors of dimension $k = 8$ each (64 raw dims $\to$ 128 via random projection). Of these 8 sub-vectors, 6 encode *shared* molecular topology (std_cos, std_sin, chi_cos, chi_sin, asym_std, asym_chi) that is identical or near-identical for enantiomers. Only 2 sub-vectors (diff_cos $= \lambda^{\text{chi}}_{\text{cos}} - \lambda^{\text{std}}_{\text{cos}}$ and diff_sin $= \lambda^{\text{chi}}_{\text{sin}} - \lambda^{\text{std}}_{\text{sin}}$) carry the chirality signal. The shared topology drowns the chirality signal during random projection, yielding enantiomer cosine similarities of 0.87–0.99.
2. **Bridged ring failure**: Molecules with bridged bicyclic systems (dextromethorphan, norbornane, camphor) fail RDKit's standard ETKDG conformer embedding, producing `ValueError` exceptions.

2 Chirality Differential Fingerprint

2.1 Construction

The differential fingerprint isolates the chirality-sensitive sub-vectors:

1. Compute standard adjacency A_{std} (bond-order weights) and chirality-modulated adjacency A_{chi} (R centers get $+w$, S centers get $-w$ on incident bonds).
2. CIP-canonical reorder both matrices.
3. Dual-helix spectral decomposition of each: eigenvalues $\lambda^{\text{std}}_{\text{cos}}[k]$, $\lambda^{\text{std}}_{\text{sin}}[k]$, $\lambda^{\text{chi}}_{\text{cos}}[k]$, $\lambda^{\text{chi}}_{\text{sin}}[k]$.
4. **Differential sub-vectors**:

$$\text{diff\_cos} = \lambda^{\text{chi}}_{\text{cos}} - \lambda^{\text{std}}_{\text{cos}} \in \mathbb{R}^k \tag{1}$$
$$\text{diff\_sin} = \lambda^{\text{chi}}_{\text{sin}} - \lambda^{\text{std}}_{\text{sin}} \in \mathbb{R}^k \tag{2}$$

5. Concatenate: $\mathbf{r} = [\text{diff\_cos}; \text{diff\_sin}] \in \mathbb{R}^{2k}$ (16 dims for $k = 8$).
6. Random projection: $\mathbf{f} = \mathbf{r} \cdot P$ where $P \in \mathbb{R}^{2k\times 128}$ (fixed seed=42).

2.2 Anti-Correlation Proof

For enantiomers with a single chiral center at CIP assignment R vs. S:

- A^R_{chi} has $+w$ modulation on bonds incident to the center.
- A^S_{chi} has $-w$ modulation on the same bonds.
- Therefore: $A^R_{\text{chi}} - A_{\text{std}} = -(A^S_{\text{chi}} - A_{\text{std}})$.
- Since eigenvalue computation is a continuous function of the adjacency: $\text{diff}(R) \approx -\text{diff}(S)$.

- After linear projection: $\mathbf{f}_R = \mathbf{r}_R \cdot P \approx -\mathbf{r}_S \cdot P = -\mathbf{f}_S$.
- Cosine similarity: $\cos(\mathbf{f}_R, \mathbf{f}_S) \approx -1.0$.

For achiral molecules, $A_{\text{chi}} = A_{\text{std}}$ (no chiral centers to modulate), so diff_cos = diff_sin = $\mathbf{0}$ and $\|\mathbf{f}\| \approx 0$.

2.3 Modes

Two modes are available via the `mode` parameter:

- `pure_diff` (default): Uses only diff_cos + diff_sin ($2k$ raw dims). Maximum anti-correlation, zero for achiral.
- `chi_diff`: Adds asym_chi $= \lambda^{\text{chi}}_{\text{cos}} - \lambda^{\text{chi}}_{\text{sin}}$ ($3k$ raw dims). Retains some structural context while still emphasizing chirality.

2.4 Benchmark: Enantiomer Cosine Similarity

Table 1: Cosine similarity between enantiomer fingerprints. Full FP (v0.6.0) vs. Differential FP (v0.7.0). All 12 pairs from Atlas v5 benchmark.

Enantiomer Pair	Full FP Cosine	Diff FP Cosine
L/D-Alanine	0.875	−0.798
L/D-Phenylalanine	0.934	−0.927
L/D-Leucine	0.944	−0.386
L/D-Valine	0.911	−0.854
L/D-Serine	0.918	−0.854
R/S-Ibuprofen	0.970	−0.740
R/S-Naproxen	0.983	−0.553
L/D-Cysteine	0.722	−0.922
L/D-Threonine	0.939	−0.089
L/D-Tryptophan	0.982	−0.484
L/D-Methionine	0.795	−0.683
Propranolol (achiral)	1.000	$\|\mathbf{f}\| = 0$
Glycine (achiral)	—	$\|\mathbf{f}\| = 0$
Ethanol (achiral)	—	$\|\mathbf{f}\| = 0$

All 11 chiral enantiomer pairs produce **negative** differential FP cosine similarity (range: −0.089 to −0.927, mean −0.663), compared to the full FP's topology-dominated positive similarities (range: 0.722 to 0.983, mean 0.907). All achiral controls produce zero-norm differential fingerprints. The weakest anti-correlation (Threonine, −0.089) is for a multi-center molecule where the two centers' differential signals partially cancel.

3 Bridged Ring Conformer Fix

3.1 Problem

Bridged bicyclic molecules—norbornane (bicyclo[2.2.1]heptane), camphor, dextromethorphan—fail RDKit's standard ETKDGv3 conformer embedding. The distance geometry algorithm cannot satisfy the distance bounds imposed by the bridged ring system, returning status −1.

In v0.6.0, the fallback chain was:

1. ETKDGv3 (fails for bridged)
2. Plain `EmbedMolecule` (fails for bridged)
3. `useRandomCoords=True` (succeeds but skips MMFF—incorrect geometry at bridgeheads)

The random-coords path skipped MMFF optimization to save time, but bridged molecules need correct 3D geometry at bridgeheads for accurate CIP stereochemistry assignment.

3.2 Detection: `_is_bridged_bicyclic()`

We detect bridged systems by examining the Smallest Set of Smallest Rings (SSSR):

1. Compute SSSR via RDKit's `GetRingInfo()`.
2. For each pair of rings, count shared atoms.
3. If any pair shares ≥ 2 atoms, the molecule contains a bridged or fused system.

This is $O(r^2 \cdot n)$ where r is the number of rings and n is the ring size—negligible for drug-like molecules.

3.3 Fallback Chain

For molecules detected as bridged, the embedding strategy is:

1. **ETKDGv3 +** `useMacrocycleTorsions`: Relaxed torsion angle constraints that help bridged geometry.
2. **ETKDGv3 +** `useBasicKnowledge=False`: Removes all knowledge-based distance bounds.
3. `useRandomCoords=True` **+ MMFF**: Random initial coordinates followed by MMFF force field optimization. Unlike the standard path, bridged molecules *always* run MMFF to ensure correct bridgehead geometry for CIP assignment.

Non-bridged molecules continue to use the original fallback chain (ETKDGv3 → plain → random coords without MMFF).

3.4 Results

Table 2: Bridged ring embedding results.

Molecule	v0.6.0	v0.7.0	MMFF
Norbornane	`ValueError` / random-coords	Success	Yes
Camphor	`ValueError` / random-coords	Success	Yes
Dextromethorphan	`ValueError`	Success	Yes
Cyclohexane (control)	Success	Success (unchanged)	Yes
Ethanol (control)	Success	Success (unchanged)	Yes

Bond length validation confirms MMFF produces physically reasonable geometries (all C–C bonds within 1.2–1.8 Å, C–H bonds within 0.8–1.3 Å).

4 Fingerprints on Result Object

The `ChiralityResult` dataclass now includes two new optional fields:

- `fingerprint: np.ndarray | None` — the 128-dim full spectral fingerprint (same as `chirality_fingerprin`
- `differential_fingerprint: np.ndarray | None` — the 128-dim chirality-only differential fingerprint.

Both are populated at all return paths in `detect()`, including the meso compound early-return. This eliminates the need for separate fingerprint computation calls in downstream ML pipelines—a single `detect()` call now returns detection results, spectral coordinates, and both fingerprint types.

5 API Reference

5.1 New Functions

```
chirality_differential_fingerprint(
    smiles_or_mol: str | Chem.Mol,
    params: HelixParams | None = None,
    nbits: int = 128,
    chiral_weight: float = 0.5,
    mode: str = "pure_diff",  # or "chi_diff"
) -> np.ndarray

batch_differential_fingerprint(
    smiles_list: list[str],
    params: HelixParams | None = None,
    nbits: int = 128,
    mode: str = "pure_diff",
    n_workers: int = 1,
) -> list[np.ndarray | None]
```

5.2 Updated Dataclass

```
@dataclass
class ChiralityResult:
    smiles: str
    chirality_score: float
    chirality_sign: float
    is_chiral: bool
    cos_eigenvalues: np.ndarray
    sin_eigenvalues: np.ndarray
    spectral_coords: SpectralCoords
    confidence: float
    fingerprint: np.ndarray | None = None          # NEW
    differential_fingerprint: np.ndarray | None = None  # NEW
```

6 Test Coverage

v0.7.0 adds 24 new tests across three files:

- **TestDifferentialFingerprint** (11 tests): output shape, deterministic, enantiomers anti-correlated, achiral near-zero norm, mode chi_diff, invalid mode, batch, camphor enantiomers, ibuprofen enantiomers.
- **TestBridgedRings** (9 tests): detection of norbornane/camphor/naphthalene, non-detection of cyclohexane/ethane, embedding of norbornane/camphor/dextromethorphan, MMFF geometry validation.
- **TestFingerprintOnResult** (5 tests): fingerprint populated on chiral/achiral/meso molecules, enantiomer differential FP anti-correlation.

Full suite: 296 passing (6 pre-existing failures in `test_targets.py` unrelated to v0.7.0 changes).

7 Conclusion

ZynerjiChirality v0.7.0 addresses the topology dilution problem identified in the v5 Atlas benchmark by introducing a chirality-only differential fingerprint that achieves anti-correlation for enantiomers by construction. The bridged ring conformer fix extends coverage to important drug scaffolds (morphinans, terpenoids, norbornane derivatives) that previously failed embedding. Both fingerprint types are now accessible directly on the detection result object, simplifying integration with downstream ML pipelines.

7.1 Future Work

- **Cross-attention conditioning**: Use differential fingerprints as conditioning vectors for generative models (replacing the failed broadcast addition approach from SMILESgpt Run 3).
- **Atlas v6**: Re-run the full 12-drug bioactivity correlation benchmark with differential fingerprints to quantify improvement in Spearman ρ.
- **Conformer ensemble differential FP**: Aggregate differential fingerprints across multiple conformers for flexible bridged molecules.

ZynerjiChirality is developed by Christian Knopp. Source code: github.com/Zynerji/ZynerjiChirality.